Approximating price series, part 2: KAN [1,1] (one spline edge) and KAN [1,h,1] (hidden layer)

Approximation study, 2020-01-02 to 2023-07-31 daily closes, three tickers. All numbers recomputed from the saved fits (approx_study/fits/*.json).

1. What is being measured

This is an in-sample APPROXIMATION task, not forecasting. Each model receives the entire close-price series of a stock, p(t), with t rescaled to [0, 1] (905 points for CBA, 901 for TSLA and LLY; prices in the file's whole-series Min-Max scale, 0 = lowest close of the period, 1 = highest), and must reproduce it as a function of t with a fixed number of parameters. The score is the RMSE and the maximum absolute error between the fitted curve and the actual series, and how both fall as the parameter budget grows (about 15, 30, 60, 120, 240). There is no train/test split and no prediction; in-sample fit says nothing about forecasting. For scale, the dotted line in the error plots is the one-day-ahead naive forecasting error of the same series (0.0201 CBA, 0.0255 TSLA, 0.0148 LLY).

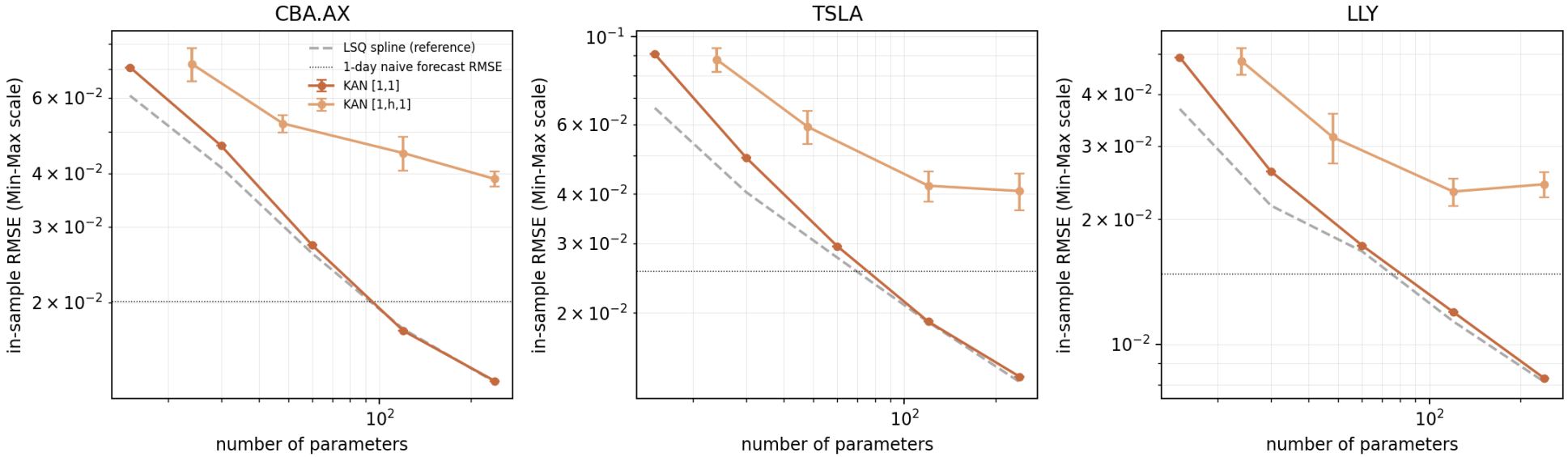
2. The two models in this document

KAN [1,1]: one edge, cubic B-spline on G intervals + silu base term
KAN [1,h,1]: hidden layer of h nodes, every edge a cubic B-spline on 3 intervals + silu
• Both are pykan 0.2.8 networks (Liu et al. 2024) fitted with L-BFGS, 80 steps, grid fixed on [0, 1], k = 3.
• Every edge computes w_b * silu(x) + w_s * spline(x); nodes only add. There are no fixed activation functions.
• KAN [1,1]: a single edge from t to the output, so the whole model is one cubic B-spline on G intervals plus the silu residual. Parameters = G + 9 (G + 3 spline coefficients, scales, base and bias terms).
• KAN [1,h,1]: the same edge function, but with a hidden layer of h nodes; 2h edges, each a cubic B-spline on only 3 intervals. Parameters about 24h. The two smallest budgets cannot be met (h = 1 gives 24 parameters).
• The one thing that changes is depth: resolution on one edge versus composition of coarse edges.
• Three seeds (0, 1, 2) were run; the [1,1] model is deterministic to 4 decimals, the [1,h,1] model is not.

3. Settings per budget

target budget	15	30	60	120
KAN [1,1] actual params	15	30	60	120
KAN [1,h,1] actual params	24	24	48	120
setting	KAN [1,1] grid G = 6; KAN [1,h,1] hidden width h = 1 (G = 3)	KAN [1,1] grid G = 21; KAN [1,h,1] hidden width h = 1 (G = 3)	KAN [1,1] grid G = 51; KAN [1,h,1] hidden width h = 2 (G = 3)	KAN [1,1] grid G

4. Error versus number of parameters (all tickers)



Points are the mean over seeds, bars the sample standard deviation across seeds (absent for deterministic models). Log-log axes: a straight line means error falls as a power of the parameter count.

Slope of log RMSE against log parameters, budgets 30-240 (how fast each model improves with size):

model	CBA.AX	TSLA	LLY	mean
KAN [1,1]	-0.61	-0.62	-0.55	-0.59
KAN [1,h,1]	-0.26	-0.34	-0.30	-0.30

5. Every fit: full table

One row per fit (ticker, model, budget, seed). RMSE and max |error| in the ticker's Min-Max scale; seconds = wall-clock fit time.

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
CBA.AX	kan11	15	0	15	0.0707	0.2201	3.1	G=6, k=3
CBA.AX	kan11	15	1	15	0.0707	0.2201	0.4	G=6, k=3
CBA.AX	kan11	15	2	15	0.0707	0.2201	0.5	G=6, k=3
CBA.AX	kan11	30	0	30	0.0464	0.1804	0.6	G=21, k=3
CBA.AX	kan11	30	1	30	0.0464	0.1804	0.5	G=21, k=3
CBA.AX	kan11	30	2	30	0.0464	0.1804	0.6	G=21, k=3
CBA.AX	kan11	60	0	60	0.0272	0.0954	1.0	G=51, k=3
CBA.AX	kan11	60	1	60	0.0272	0.0954	0.8	G=51, k=3
CBA.AX	kan11	60	2	60	0.0272	0.0954	0.9	G=51, k=3
CBA.AX	kan11	120	0	120	0.0172	0.0804	1.3	G=111, k=3
CBA.AX	kan11	120	1	120	0.0172	0.0804	1.1	G=111, k=3
CBA.AX	kan11	120	2	120	0.0172	0.0804	1.0	G=111, k=3
CBA.AX	kan11	240	0	240	0.0131	0.0696	4.4	G=231, k=3
CBA.AX	kan11	240	1	240	0.0131	0.0696	1.8	G=231, k=3
CBA.AX	kan11	240	2	240	0.0131	0.0696	1.6	G=231, k=3
CBA.AX	kan1h1	15	0	24	0.0757	0.2444	1.4	h=1, G=3, k=3
CBA.AX	kan1h1	15	1	24	0.0646	0.1943	1.8	h=1, G=3, k=3
CBA.AX	kan1h1	15	2	24	0.0757	0.2443	1.3	h=1, G=3, k=3
CBA.AX	kan1h1	30	0	24	0.0757	0.2444	1.5	h=1, G=3, k=3
CBA.AX	kan1h1	30	1	24	0.0646	0.1943	1.9	h=1, G=3, k=3
CBA.AX	kan1h1	30	2	24	0.0757	0.2443	1.6	h=1, G=3, k=3
CBA.AX	kan1h1	60	0	48	0.0513	0.1968	5.2	h=2, G=3, k=3
CBA.AX	kan1h1	60	1	48	0.0551	0.2122	5.7	h=2, G=3, k=3
CBA.AX	kan1h1	60	2	48	0.0504	0.1437	5.8	h=2, G=3, k=3
CBA.AX	kan1h1	120	0	120	0.0447	0.1634	7.2	h=5, G=3, k=3
CBA.AX	kan1h1	120	1	120	0.0486	0.1974	6.7	h=5, G=3, k=3

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
CBA.AX	kan1h1	120	2	120	0.0404	0.1749	6.6	h=5, G=3, k=3
CBA.AX	kan1h1	240	0	240	0.0372	0.1407	9.6	h=10, G=3, k=3
CBA.AX	kan1h1	240	1	240	0.0390	0.1371	9.9	h=10, G=3, k=3
CBA.AX	kan1h1	240	2	240	0.0403	0.1464	9.6	h=10, G=3, k=3
TSLA	kan11	15	0	15	0.0905	0.3084	0.4	G=6, k=3
TSLA	kan11	15	1	15	0.0905	0.3084	0.5	G=6, k=3
TSLA	kan11	15	2	15	0.0905	0.3084	0.5	G=6, k=3
TSLA	kan11	30	0	30	0.0495	0.1920	0.5	G=21, k=3
TSLA	kan11	30	1	30	0.0495	0.1920	0.6	G=21, k=3
TSLA	kan11	30	2	30	0.0495	0.1920	0.5	G=21, k=3
TSLA	kan11	60	0	60	0.0295	0.1393	0.9	G=51, k=3
TSLA	kan11	60	1	60	0.0295	0.1393	0.7	G=51, k=3
TSLA	kan11	60	2	60	0.0295	0.1393	0.7	G=51, k=3
TSLA	kan11	120	0	120	0.0190	0.0818	1.9	G=111, k=3
TSLA	kan11	120	1	120	0.0190	0.0818	1.2	G=111, k=3
TSLA	kan11	120	2	120	0.0190	0.0818	1.0	G=111, k=3
TSLA	kan11	240	0	240	0.0138	0.0721	1.9	G=231, k=3
TSLA	kan11	240	1	240	0.0138	0.0721	2.0	G=231, k=3
TSLA	kan11	240	2	240	0.0138	0.0721	1.9	G=231, k=3
TSLA	kan1h1	15	0	24	0.0912	0.3136	1.7	h=1, G=3, k=3
TSLA	kan1h1	15	1	24	0.0806	0.2465	2.5	h=1, G=3, k=3
TSLA	kan1h1	15	2	24	0.0912	0.3136	1.4	h=1, G=3, k=3
TSLA	kan1h1	30	0	24	0.0912	0.3136	2.3	h=1, G=3, k=3
TSLA	kan1h1	30	1	24	0.0806	0.2465	3.1	h=1, G=3, k=3
TSLA	kan1h1	30	2	24	0.0912	0.3136	1.4	h=1, G=3, k=3
TSLA	kan1h1	60	0	48	0.0542	0.2132	5.9	h=2, G=3, k=3
TSLA	kan1h1	60	1	48	0.0580	0.2421	6.2	h=2, G=3, k=3
TSLA	kan1h1	60	2	48	0.0654	0.2188	3.9	h=2, G=3, k=3

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
TSLA	kan1h1	120	0	120	0.0460	0.1827	7.6	h=5, G=3, k=3
TSLA	kan1h1	120	1	120	0.0388	0.1808	7.4	h=5, G=3, k=3
TSLA	kan1h1	120	2	120	0.0412	0.1816	7.2	h=5, G=3, k=3
TSLA	kan1h1	240	0	240	0.0433	0.1921	10.3	h=10, G=3, k=3
TSLA	kan1h1	240	1	240	0.0431	0.1828	10.3	h=10, G=3, k=3
TSLA	kan1h1	240	2	240	0.0357	0.1781	10.7	h=10, G=3, k=3
LLY	kan11	15	0	15	0.0491	0.1440	0.5	G=6, k=3
LLY	kan11	15	1	15	0.0491	0.1440	0.5	G=6, k=3
LLY	kan11	15	2	15	0.0491	0.1440	0.5	G=6, k=3
LLY	kan11	30	0	30	0.0261	0.0790	0.6	G=21, k=3
LLY	kan11	30	1	30	0.0261	0.0790	1.7	G=21, k=3
LLY	kan11	30	2	30	0.0261	0.0790	0.6	G=21, k=3
LLY	kan11	60	0	60	0.0173	0.0635	0.8	G=51, k=3
LLY	kan11	60	1	60	0.0173	0.0635	0.8	G=51, k=3
LLY	kan11	60	2	60	0.0173	0.0635	0.7	G=51, k=3
LLY	kan11	120	0	120	0.0120	0.0483	1.1	G=111, k=3
LLY	kan11	120	1	120	0.0120	0.0483	1.2	G=111, k=3
LLY	kan11	120	2	120	0.0120	0.0483	1.3	G=111, k=3
LLY	kan11	240	0	240	0.0083	0.0342	1.5	G=231, k=3
LLY	kan11	240	1	240	0.0083	0.0342	2.6	G=231, k=3
LLY	kan11	240	2	240	0.0083	0.0342	1.7	G=231, k=3
LLY	kan1h1	15	0	24	0.0473	0.1467	4.1	h=1, G=3, k=3
LLY	kan1h1	15	1	24	0.0521	0.1653	1.8	h=1, G=3, k=3
LLY	kan1h1	15	2	24	0.0450	0.1297	2.8	h=1, G=3, k=3
LLY	kan1h1	30	0	24	0.0473	0.1467	3.9	h=1, G=3, k=3
LLY	kan1h1	30	1	24	0.0521	0.1653	1.6	h=1, G=3, k=3
LLY	kan1h1	30	2	24	0.0450	0.1297	2.7	h=1, G=3, k=3
LLY	kan1h1	60	0	48	0.0289	0.0887	5.5	h=2, G=3, k=3

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
LLY	kan1h1	60	1	48	0.0291	0.0918	6.4	h=2, G=3, k=3
LLY	kan1h1	60	2	48	0.0365	0.1197	6.2	h=2, G=3, k=3
LLY	kan1h1	120	0	120	0.0235	0.0791	7.3	h=5, G=3, k=3
LLY	kan1h1	120	1	120	0.0215	0.0736	8.3	h=5, G=3, k=3
LLY	kan1h1	120	2	120	0.0250	0.0901	6.6	h=5, G=3, k=3
LLY	kan1h1	240	0	240	0.0229	0.0717	9.3	h=10, G=3, k=3
LLY	kan1h1	240	1	240	0.0261	0.0856	11.4	h=10, G=3, k=3
LLY	kan1h1	240	2	240	0.0238	0.0750	11.6	h=10, G=3, k=3

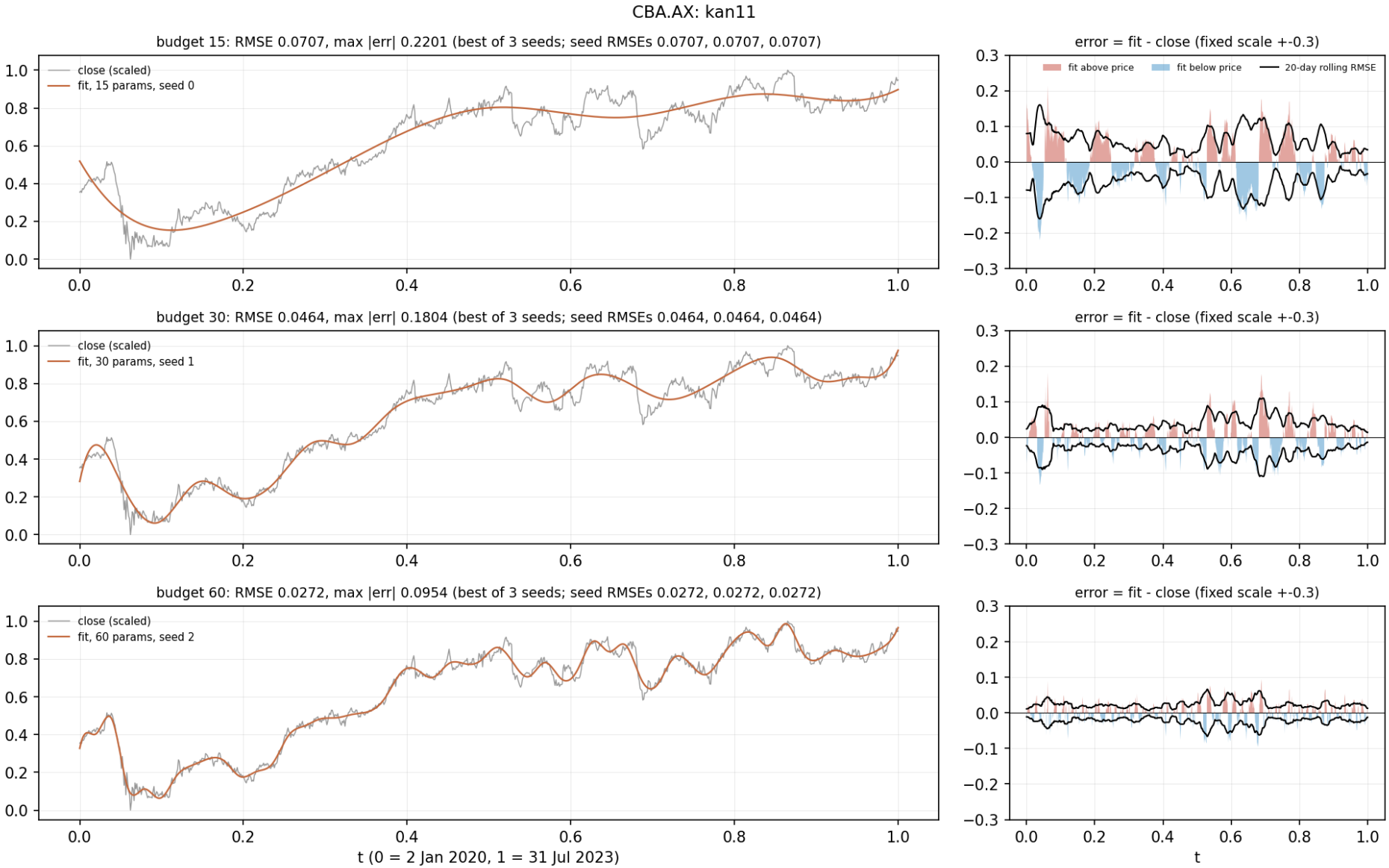
6. Summary per ticker and budget (mean +- sd over seeds)

ticker	budget	KAN [1,1] RMSE	max |err|	params	KAN [1,h,1] RMSE	max |err|	params
CBA.AX	15	0.0707 +- 0.0000	0.2201	15	0.0720 +- 0.0064	0.2277	24
CBA.AX	30	0.0464 +- 0.0000	0.1804	30	0.0720 +- 0.0064	0.2277	24
CBA.AX	60	0.0272 +- 0.0000	0.0954	60	0.0523 +- 0.0025	0.1842	48
CBA.AX	120	0.0172 +- 0.0000	0.0804	120	0.0446 +- 0.0041	0.1786	120
CBA.AX	240	0.0131 +- 0.0000	0.0696	240	0.0388 +- 0.0016	0.1414	240
TSLA	15	0.0905 +- 0.0000	0.3084	15	0.0877 +- 0.0061	0.2912	24
TSLA	30	0.0495 +- 0.0000	0.1920	30	0.0877 +- 0.0061	0.2912	24
TSLA	60	0.0295 +- 0.0000	0.1393	60	0.0592 +- 0.0057	0.2247	48
TSLA	120	0.0190 +- 0.0000	0.0818	120	0.0420 +- 0.0037	0.1817	120
TSLA	240	0.0138 +- 0.0000	0.0721	240	0.0407 +- 0.0044	0.1843	240
LLY	15	0.0491 +- 0.0000	0.1440	15	0.0481 +- 0.0036	0.1472	24
LLY	30	0.0261 +- 0.0000	0.0790	30	0.0481 +- 0.0036	0.1472	24
LLY	60	0.0173 +- 0.0000	0.0635	60	0.0315 +- 0.0043	0.1001	48
LLY	120	0.0120 +- 0.0000	0.0483	120	0.0233 +- 0.0018	0.0809	120
LLY	240	0.0083 +- 0.0000	0.0342	240	0.0243 +- 0.0017	0.0774	240

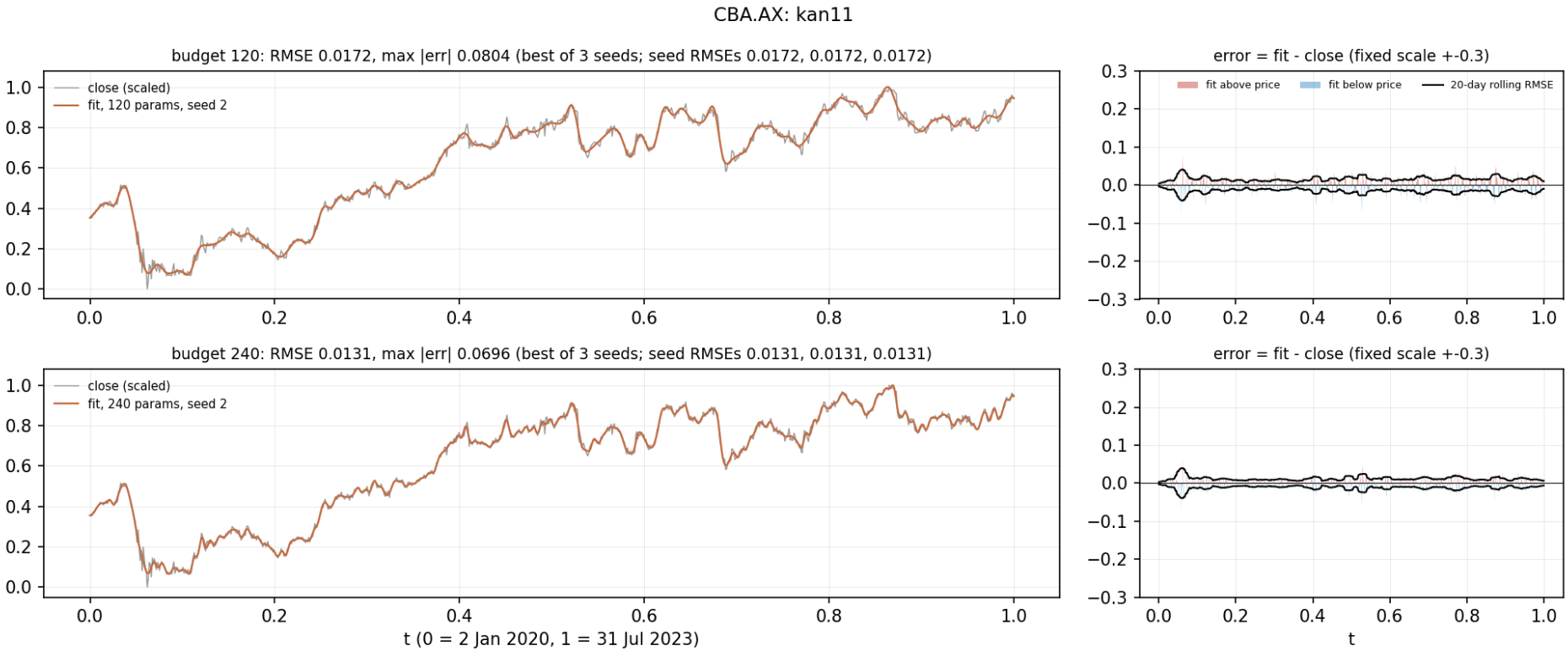
7. The fits themselves (following pages)

For every ticker, model and budget: left, the fitted curve over the actual series (best seed where there are seeds; all seed RMSEs in the title); right, the error fit - close on a fixed +-0.3 scale so panels are comparable: red shading where the fit sits above the price, blue where below, and the black envelope is the 20-day rolling RMSE (+-), i.e. the local size of the error with the day-to-day jitter smoothed out. Vertical lines in spline panels are knot positions.

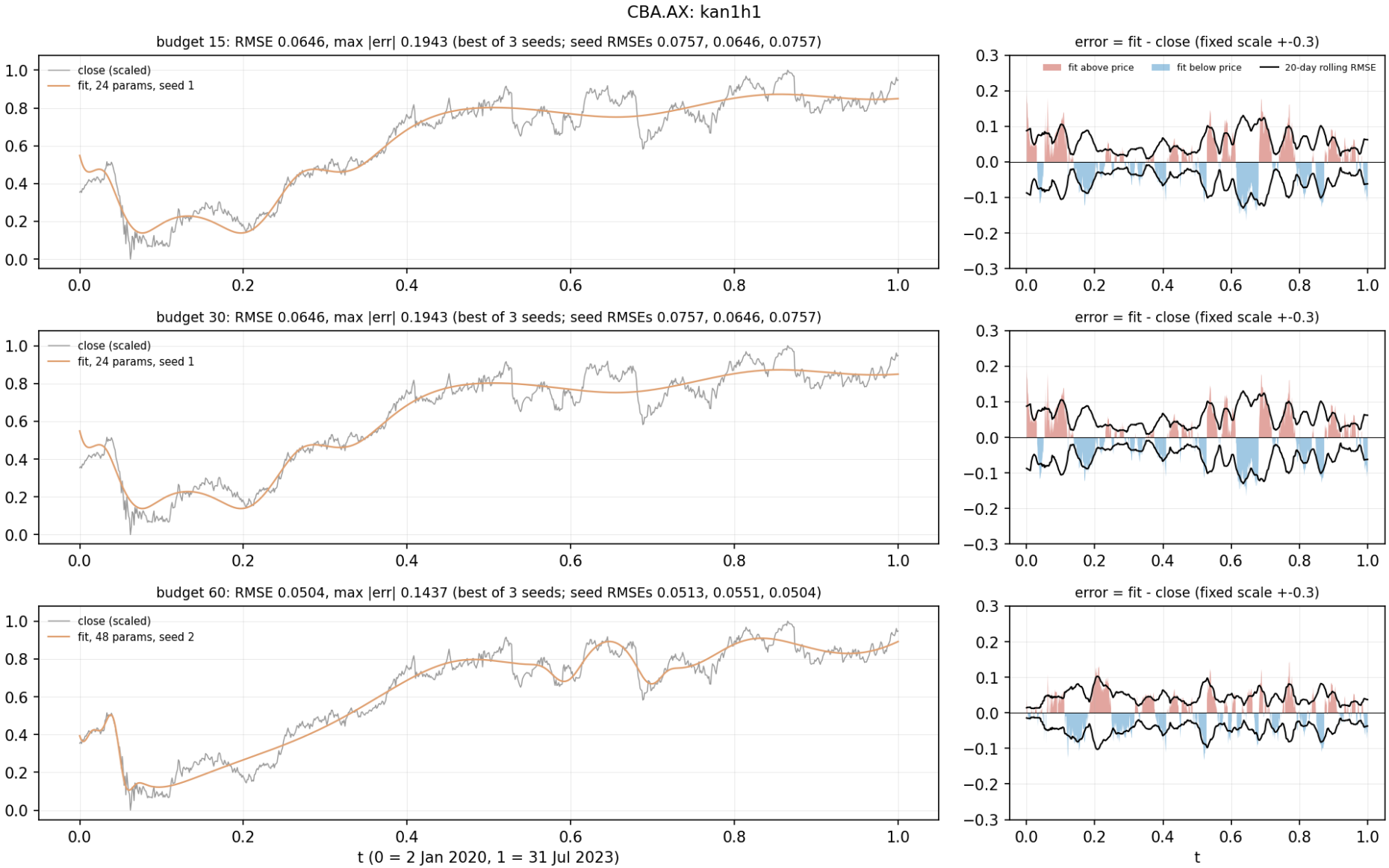
CBA.AX (Commonwealth Bank, benchmark series), KAN [1,1]: one edge, cubic B-spline on G intervals + silu base term (budgets 15, 30, 60)



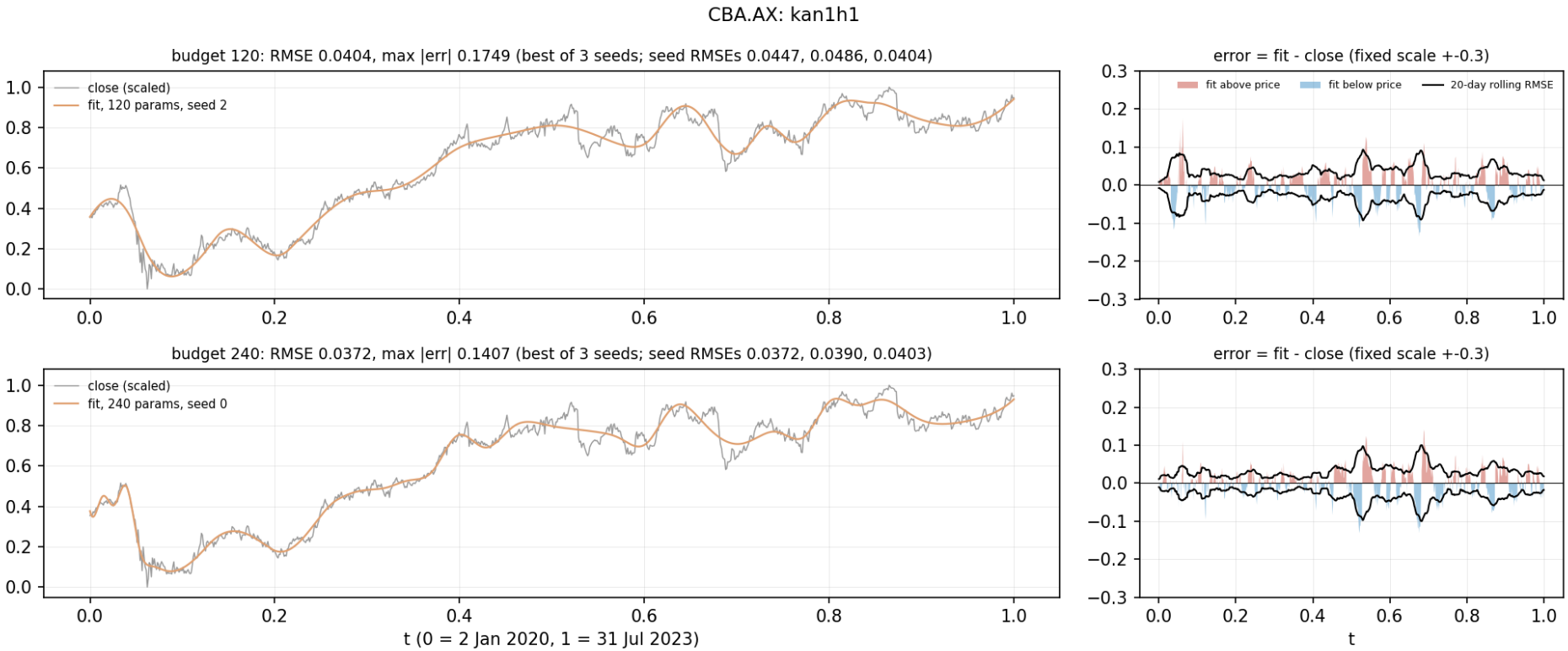
CBA.AX (Commonwealth Bank, benchmark series), KAN [1,1]: one edge, cubic B-spline on G intervals + silu base term (budgets 120, 240)



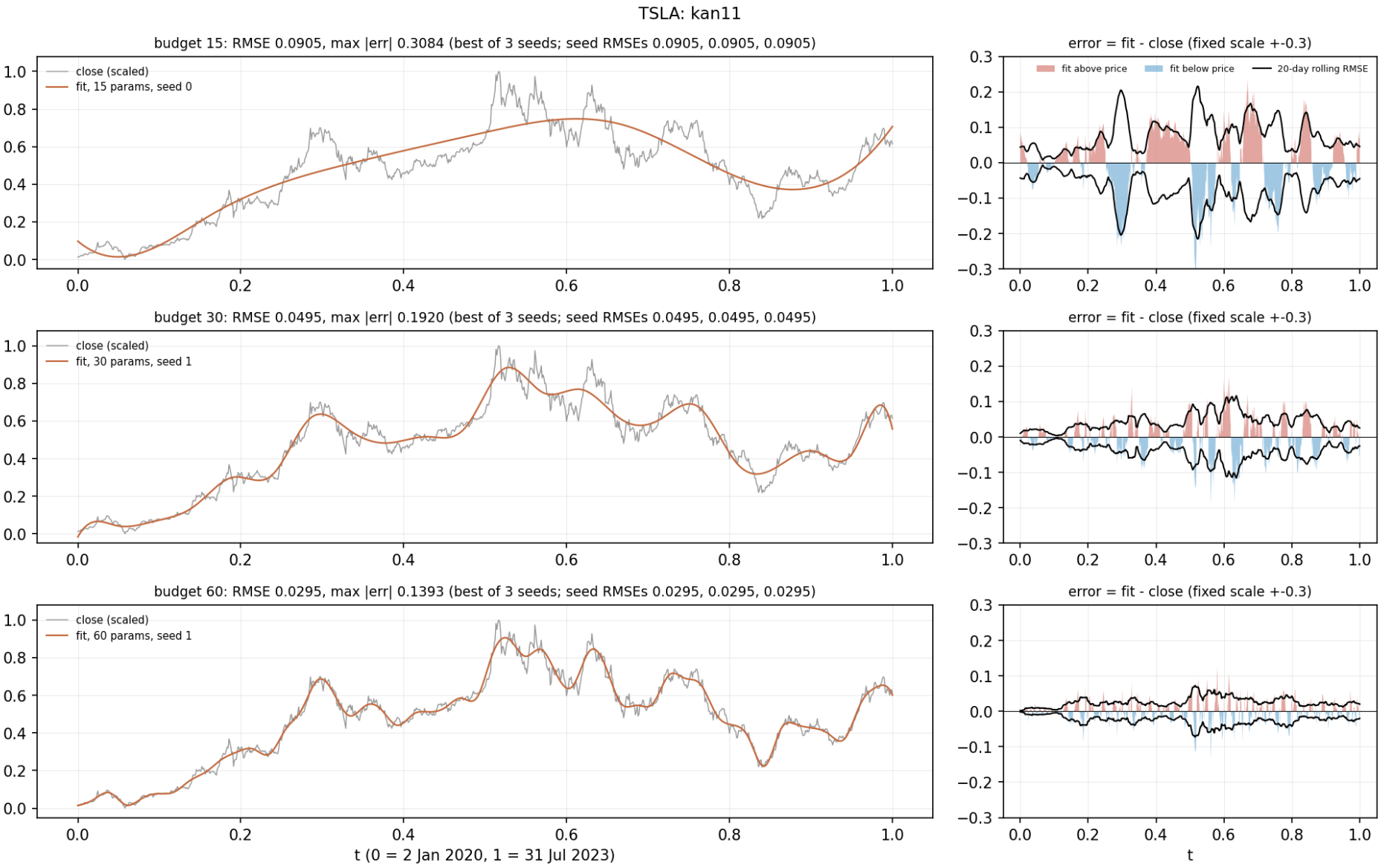
CBA.AX (Commonwealth Bank, benchmark series), KAN [1,h,1]: hidden layer of h nodes, every edge a cubic B-spline on 3 intervals + silu (budgets 15, 30, 60)



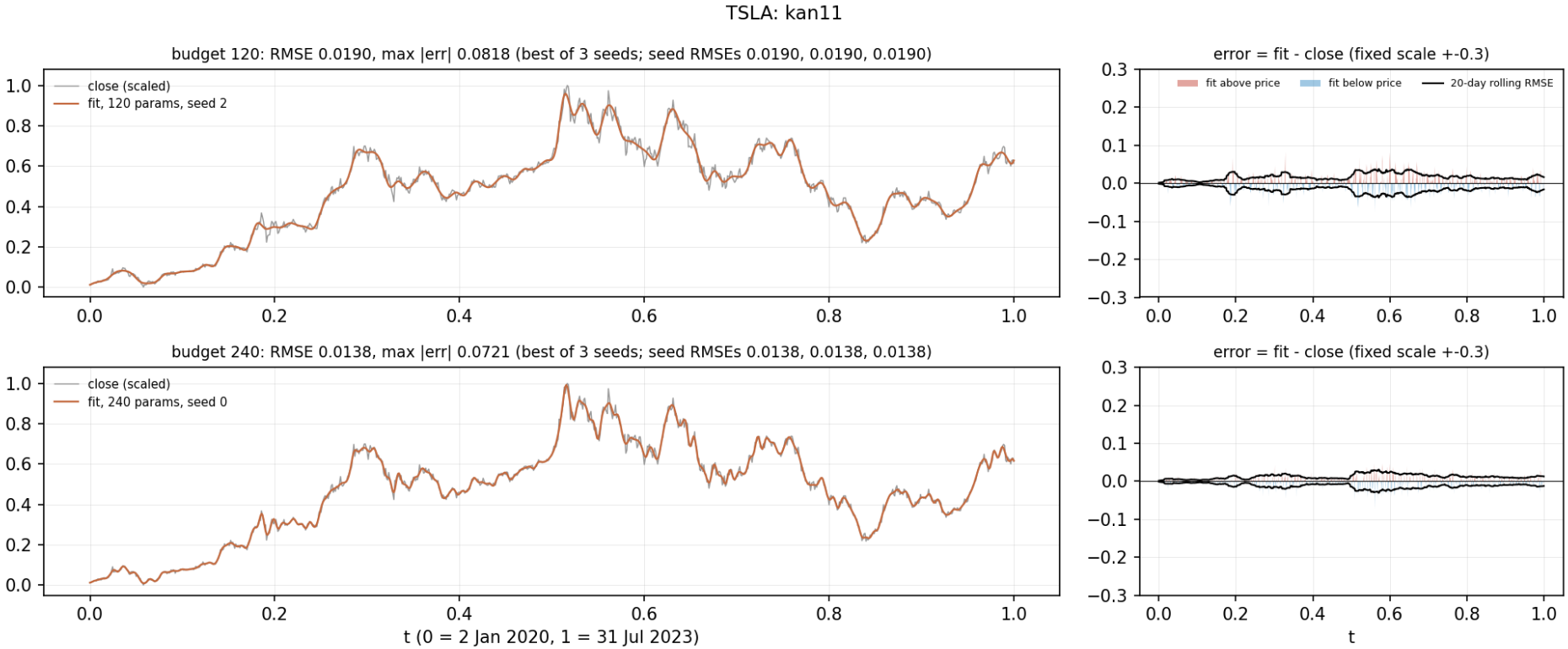
CBA.AX (Commonwealth Bank, benchmark series), KAN [1,h,1]: hidden layer of h nodes, every edge a cubic B-spline on 3 intervals + silu (budgets 120, 240)



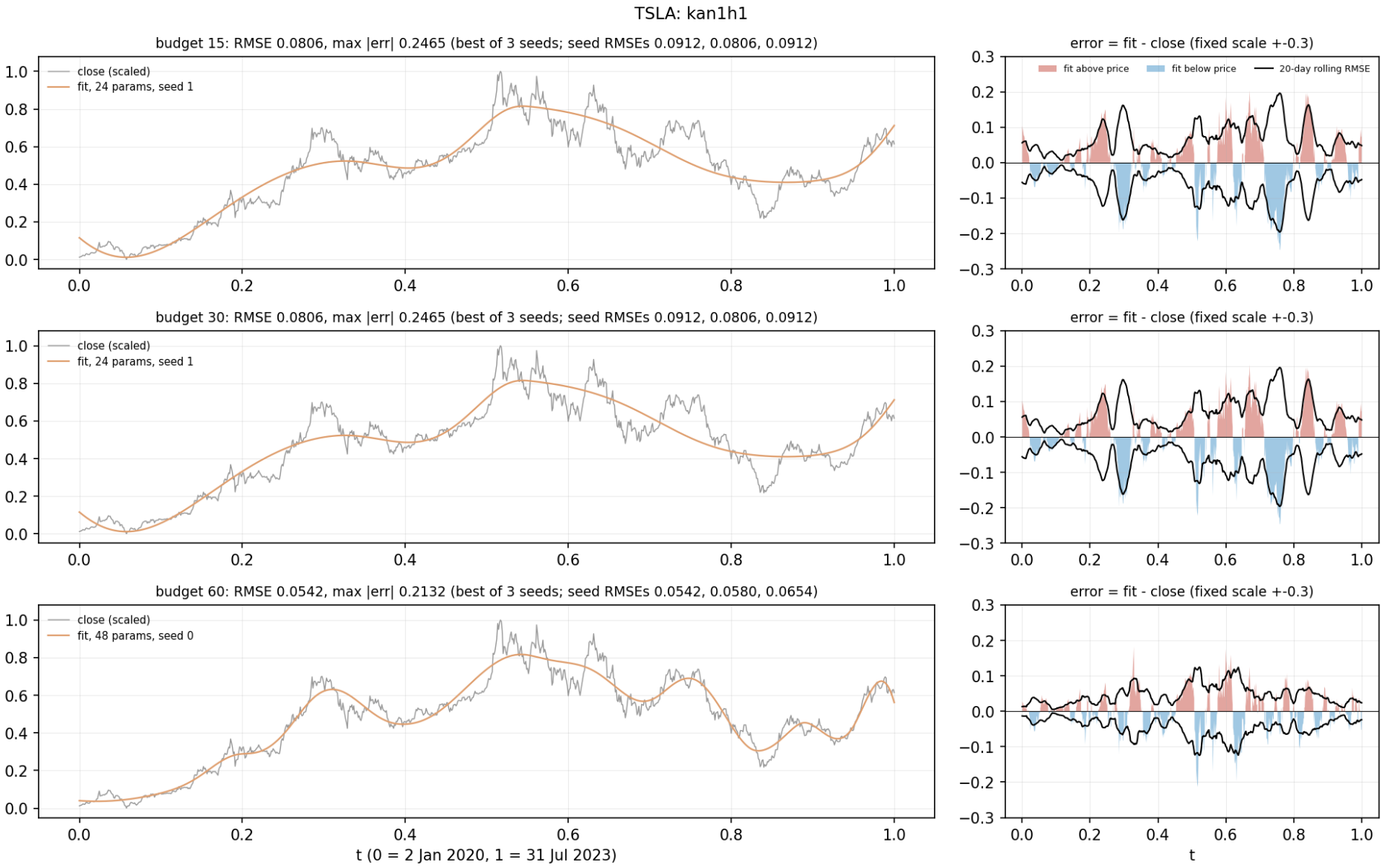
TSLA (Tesla, volatile), KAN [1,1]: one edge, cubic B-spline on G intervals + silu base term (budgets 15, 30, 60)



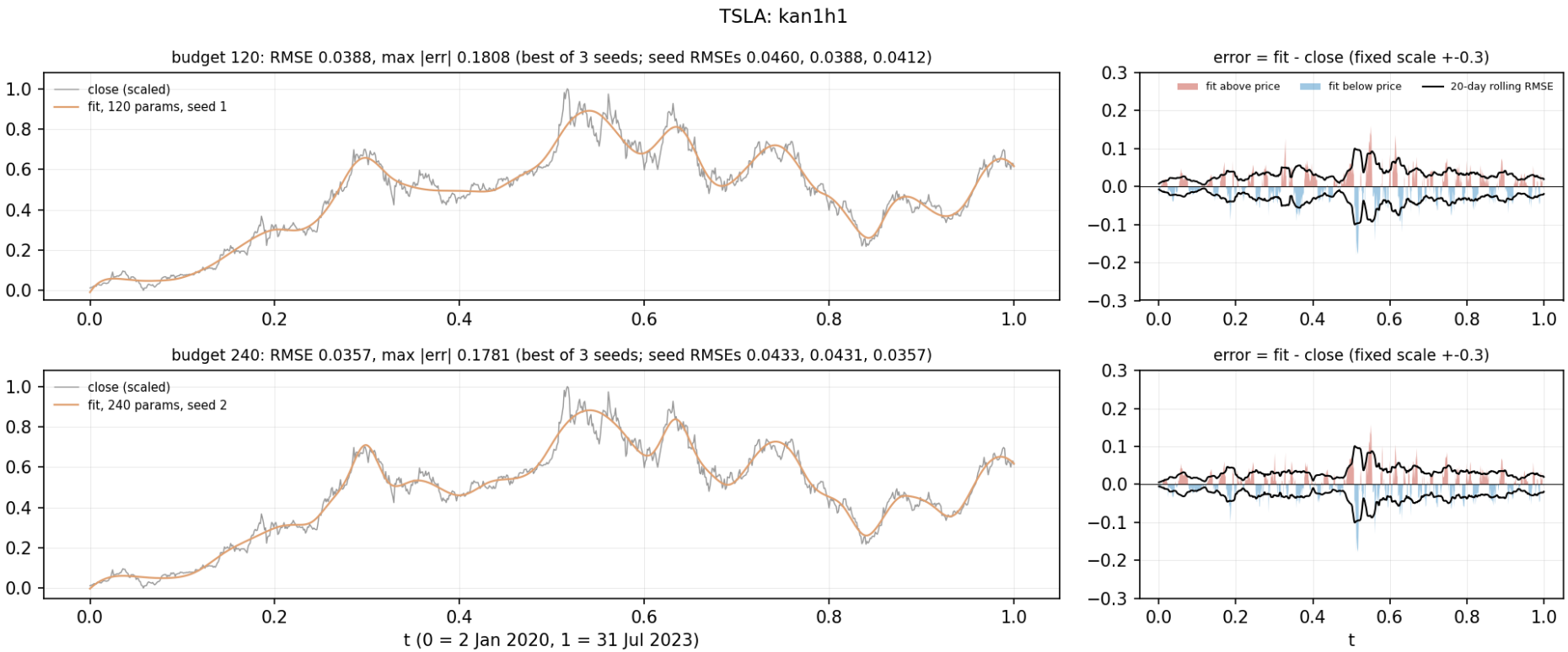
TSLA (Tesla, volatile), KAN [1,1]: one edge, cubic B-spline on G intervals + silu base term (budgets 120, 240)



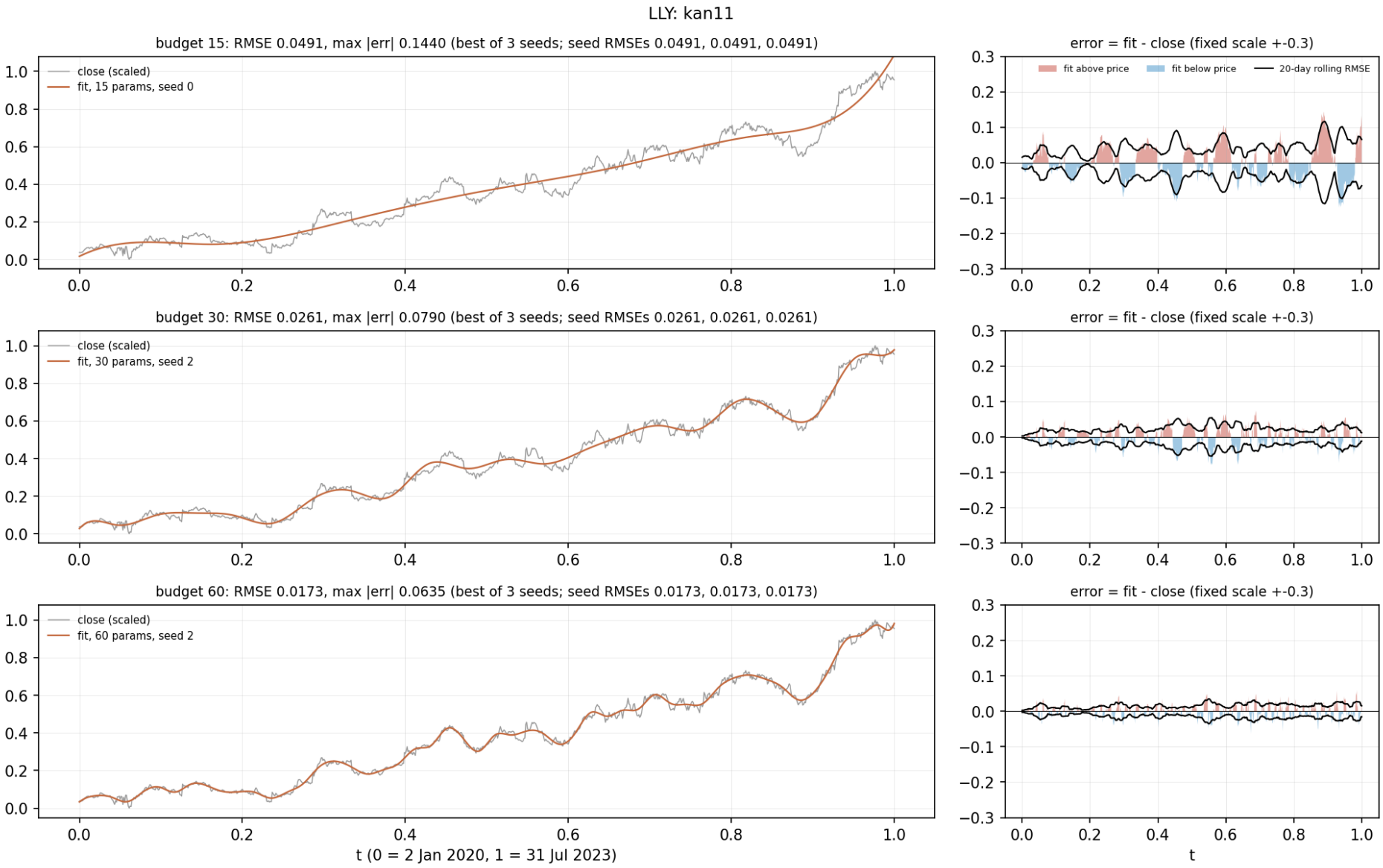
TSLA (Tesla, volatile), KAN [1,h,1]: hidden layer of h nodes, every edge a cubic B-spline on 3 intervals + silu (budgets 15, 30, 60)



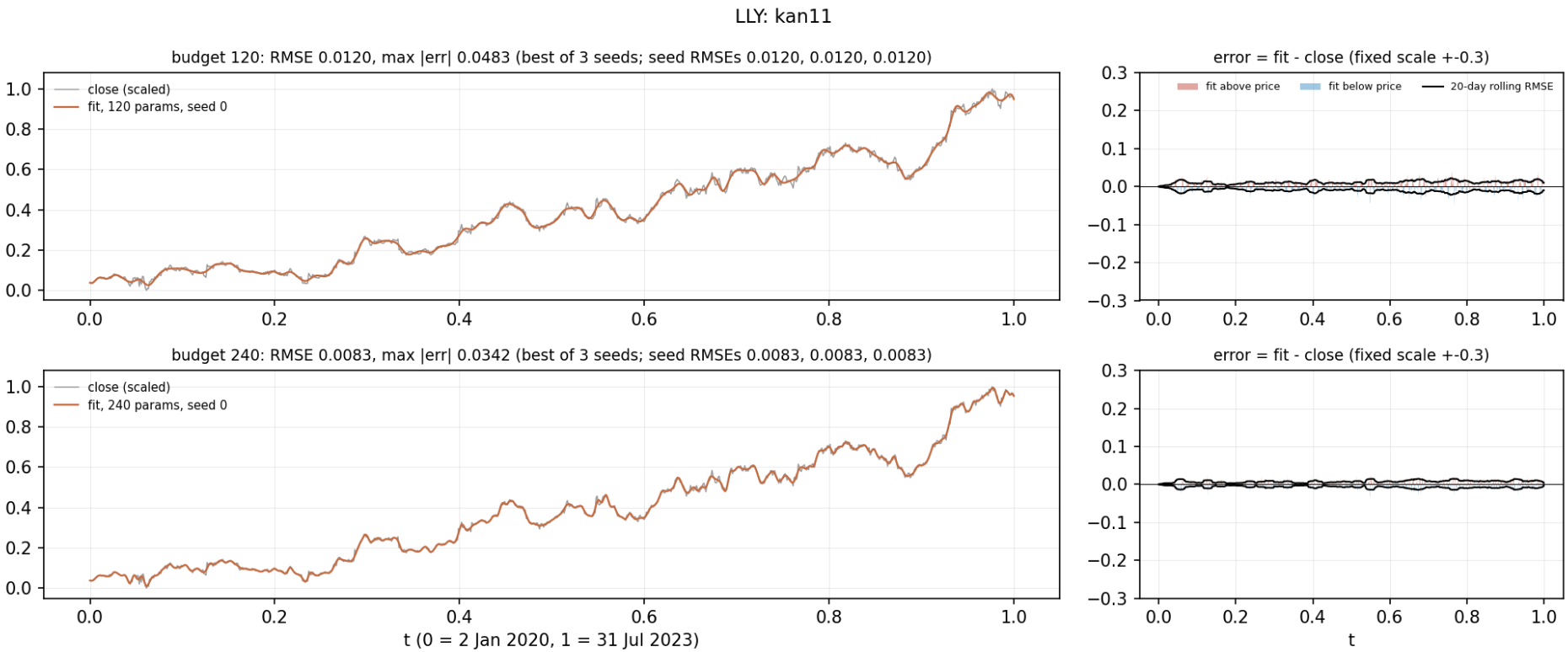
TSLA (Tesla, volatile), KAN [1,h,1]: hidden layer of h nodes, every edge a cubic B-spline on 3 intervals + silu (budgets 120, 240)



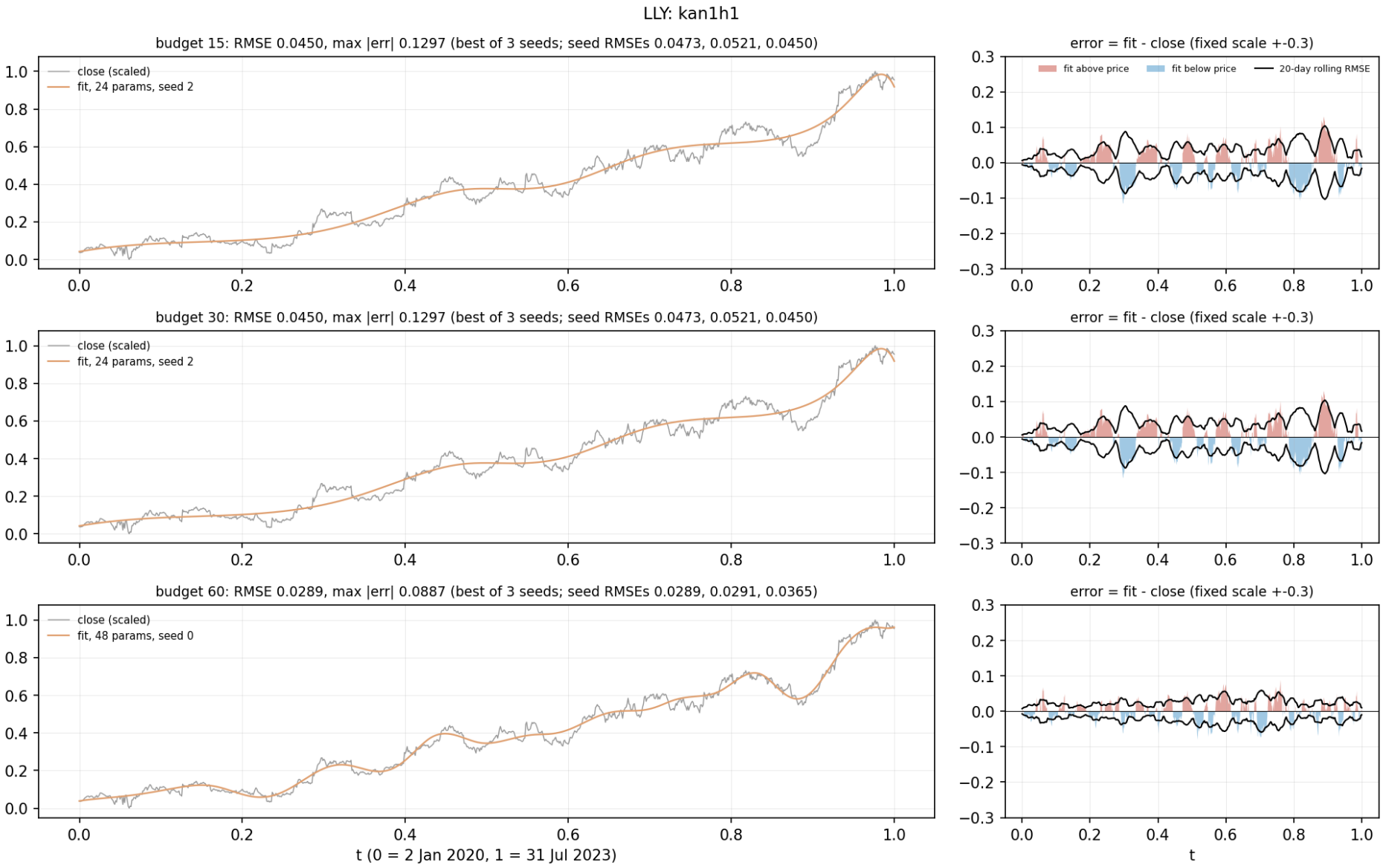
LLY (Eli Lilly, stable trend), KAN [1,1]: one edge, cubic B-spline on G intervals + silu base term (budgets 15, 30, 60)



LLY (Eli Lilly, stable trend), KAN [1,1]: one edge, cubic B-spline on G intervals + silu base term (budgets 120, 240)



LLY (Eli Lilly, stable trend), KAN [1,h,1]: hidden layer of h nodes, every edge a cubic B-spline on 3 intervals + silu (budgets 15, 30, 60)



LLY (Eli Lilly, stable trend), KAN [1,h,1]: hidden layer of h nodes, every edge a cubic B-spline on 3 intervals + silu (budgets 120, 240)

